Mouhanned Mejri

Computer Science Engineering Student | Software Engineering | Generative AI

Tunis, Tunisia | Mouhannedmejri10@gmail.com | LinkedIn | GitHub | Codeforces Specialist (≈1500)

Profile

Computer Science Engineering student at the Higher National Engineering School of Tunis (ENSIT) with a strong foundation in software engineering, algorithms, and full-stack development. Hands-on experience building AI-powered applications, LLM/RAG pipelines, multi-agent systems, web platforms, and mobile applications. Interested in Generative AI, AI agents, intelligent automation, and accessible human-computer interaction.

Technical Skills

Programming: Python, C++, C, Java, JavaScript, TypeScript
Generative AI & AI: LLMs, RAG, Prompt Engineering, Multimodal AI, AI Assistants, NLP, AI Agents, Agentic Workflows
Frameworks & Libraries: FastAPI, Flask, Spring Boot, React, React Native, Next.js, Angular, Ionic, Expo
Web & Backend: REST APIs, WebView, Chrome Extensions, DOM Manipulation, Browser Automation
Database & Authentication: PostgreSQL, SQL, Prisma, Supabase
Tools & Platforms: Git, GitHub, Docker, Uvicorn, Plasmo
Competitive Programming: Algorithms, Data Structures, Problem Solving, Codeforces Specialist (≈1500)

Professional Experience

Software Development Intern Summer 2026
Talan Summer Camp *Tunisia*

- Developed **Navi**, an AI-powered accessibility ecosystem that automatically detects user interaction difficulties, adapts interfaces, and executes web navigation tasks through conversational AI.
- Built a **multi-agent architecture** with specialized agents for user profiling, conversational intent routing, DOM analysis, and autonomous browser planning and execution.
- Developed cross-platform clients using **React Native, Expo, and Plasmo**, combining a mobile application with a Chrome MV3 browser extension.
- Implemented backend services with **FastAPI and Python**, integrating LLM-powered reasoning, multimodal image understanding, speech interaction, page analysis, and browser automation.
- Integrated **Mistral/Pixtral, Groq, Gemini, Voxtral, and Firecrawl** for multimodal understanding, planning, speech interaction, and web content processing.

Software Development Intern 2025
Krenova Consulting *Tunisia*

- Designed and developed an end-to-end e-commerce platform specialized in automotive spare parts.
- Integrated an **AI-powered recommendation assistant** to suggest suitable components based on vehicle specifications.
- Contributed to frontend development using **React, TypeScript, Tailwind CSS, and modern component-based architecture**.

Webmaster 2024 – 2025
IEEE ENSIT Student Branch *Tunisia*

- Maintained and developed features for the official IEEE ENSIT Student Branch website.
- Led and coordinated the branch development team while managing feature implementation and technical tasks.

Project Manager 2024 – 2025
IEEE ENSIT Computer Society Chapter *Tunisia*

- Managed project lifecycles including requirements analysis, task allocation, milestone tracking, and quality assurance.
- Facilitated communication and coordination across development teams to ensure timely project delivery.

Programme Manager 2025
TomorrowLand 2.0 IT Event *Tunisia*

- Structured workshops, competitions, and activities into a coherent technical event programme while coordinating participant flow and scheduling.

Selected Projects

Navi – AI Accessibility Assistant *2026*

- Built an intelligent accessibility ecosystem designed to adapt digital interfaces to users through **automatic detection, dynamic UI adaptation, and agentic task automation**.
- Implemented passive interaction monitoring using gesture and behavioral telemetry including pinch-zoom, tap misses, rapid double-taps, scroll hesitation, and mistaps to identify potential user struggles.
- Developed a two-layer adaptation system capable of dynamically modifying web pages and mobile interfaces through CSS/DOM transformations and personalized capability profiles.

- Implemented specialized AI agents for **user profiling, conversational interaction, DOM/page analysis, and browser planning and execution**.
- Enabled natural-language automation of complex web workflows such as navigation, clicking, typing, scrolling, and page summarization.
- **Technologies:** React Native, Expo, Plasmo, FastAPI, Python, Supabase, Mistral/Pixtral, Groq, Gemini, Voxtral, Firecrawl

BugHunting – AI-Assisted Bug Bounty Platform *2026*

- Engineered a security platform connecting vulnerability researchers with clients seeking security assessments for defined project scopes.
- Integrated a **RAG/LLM-based classification pipeline** to automatically identify and filter spam, fraudulent, and low-quality vulnerability reports.
- Implemented role-based workflows and a severity-tiered bounty system for managing vulnerability validation and researcher rewards.
- **Technologies:** Next.js, TypeScript, PostgreSQL, Prisma, Python, RAG, LLMs

Apex Arena – Competitive Programming Platform *2026*

- Developed a gamified competitive programming platform supporting team-based contests and structured tournament brackets.
- Implemented a dual-tournament system with Champions and Lower divisions, head-to-head matches, rankings, and promotion/relegation mechanics.
- Designed a competitive workflow combining algorithmic problem solving, tournament progression, and team rankings.

Enchère Express – Online Auction Platform *2025*

- Built a real-time auction platform supporting multiple asset categories including fine art, real estate, and vehicles.
- Implemented authenticated bidding, configurable auction windows, automated closing, and winner notifications.

FreelanceHub – Freelance Marketplace Mobile Application *2025*

- Developed a cross-platform mobile marketplace connecting clients with freelance professionals.
- Implemented project posting, freelancer proposals, bidding workflows, and real-time in-app messaging.

Education

Higher National Engineering School of Tunis (ENSIT) 2024 – Present
Computer Science Engineering Degree *Tunis, Tunisia*

Languages

Arabic: Native **French:** Fluent **English:** Fluent **German:** Basic Proficiency